

4. $f(x) = \sqrt[3]{x^2}$

13. $f(x) = \frac{2x^4 - 3x + 1}{x+1}$

4. $f(x) = \sqrt[3]{x^2} \Rightarrow f'(x) = \frac{2x^{-\frac{1}{3}}}{3}$
 $= \frac{2}{3\sqrt[3]{x}}$

13. $f(x) = \frac{2x^4}{x+1} - \frac{3x}{x+1} + \frac{1}{x+1}$

$\Rightarrow f'(x) = \frac{8x^3(x+1) - 2x^4}{(x+1)^2} - \frac{3(x+1) - 3x}{(x+1)^2} + \frac{-1}{(x+1)^2}$
 $= \frac{8x^3(x+1) - 2x^4 - (3x+1) - 3x - 1}{(x+1)^2}$

21. $f(x) = x^2 \sin x \Rightarrow f'(x) = 2x \sin(x) + x^2 \cos(x)$

22. $f(x) = (x^2+1) \tan(x) \Rightarrow f'(x) = 2x \tan(x) + (x^2+1) \sec^2(x)$

25. $f(x) = \frac{\sin(x)}{x^2+1} \Rightarrow f'(x) = \frac{\cos(x)(x^2+1) - \sin(x)(2x)}{(x^2+1)^2}$
 $= \frac{\cos(x)(x^2+1) - 2x \sin(x)}{(x^2+1)^2}$

28. $f(x) = 3xe^{3x+5}$

$\Rightarrow f'(x) = 3e^{3x+5} + 3x \cdot 3e^{3x+5} = 3e^{3x+5}(1+3x)$

29. $f(x) = \ln(x^2) - e^{5x} \Rightarrow f'(x) = \frac{1}{x^2} \cdot 2x - e^{5x} \cdot 5 = \frac{2x}{x^2} - 5e^{5x}$

30. $f(x) = e^{2x} \cdot \sin(x) \Rightarrow f'(x) = 2e^{2x} \sin(x) + e^{2x} \cos(x)$

31. $f(x) = \sin(x^2 - 3x)$

$f'(x) = \cos(x^2 - 3x) \cdot (2x - 3)$

33. $f(x) = (3x^2 - 5)^{\frac{2}{3}} \Rightarrow f'(x) = \frac{2}{3} (3x^2 - 5)^{-\frac{1}{3}} \cdot (6x) = \frac{4x}{\sqrt[3]{3x^2 - 5}}$

34. $f(x) = \frac{\sin(x+2)}{3x^2} \Rightarrow f'(x) = \frac{\cos(x+2)(3x^2) - \sin(x+2)(6x)}{(3x^2)^2} = \frac{3x^2 \cos(x+2) - 2 \sin(x+2)x}{9x^4}$

28. $f(x) = 3xe^{3x+5}$

29. $f(x) = \ln(x^2) - e^{5x}$

30. $f(x) = e^{2x} \cdot \sin(x)$

31. $f(x) = \sin(x^2 - 3x)$

33. $f(x) = \sqrt[3]{(3x^2 - 5)^2}$

34. $f(x) = \frac{\sin(x+2)}{3x^2}$

$f(x) = 2^{2x} \Rightarrow f \circ g \Rightarrow 2^x$
 $g \Rightarrow 2x$

$\frac{\sin(x)}{x^2}$

Cál Int 5/8 Prof. Horacio

Propiedades de la derivada: Sean $f, g: [a, b] \rightarrow \mathbb{R}$

funciones derivables y $\alpha \in \mathbb{R}$

$$i) (\alpha f(x))' = \alpha f'(x)$$

$$ii) (f(x) \pm g(x))' = f'(x) \pm g'(x)$$

$$iii) (f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$* (fgh)' = f'gh + fg'h + fgh'$$

$$iv) \left(f(x) \frac{1}{g(x)} \right)' = \left(\frac{f(x)}{g(x)} \right)' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}, \quad g(x) \neq 0$$

$$f(x) = \frac{2x^4 - 3x + 1}{x} \Rightarrow f'(x) = 6x^2 + -x^{-2}$$

$$f(x) = \frac{2x^4}{x} - \frac{3x}{x} + \frac{1}{x}$$

$$= 2x^3 - 3 + \frac{1}{x}$$

$$f'(x) = \frac{(8x^3 - 3) \cdot x - (2x^4 - 3x + 1)}{x^2}$$

$$= \frac{8x^4 - 3x - 2x^4 + 3x - 1}{x^2}$$

$$= \frac{6x^4 - 1}{x^2}$$

$$= 6x^2 - \frac{1}{x^2}$$

$$b) f(x) = \frac{(2x^2 + 1) \sin(x)}{x^2 + 1}$$

$$f'(x) = \frac{(4x \sin(x) + (2x^2 + 1) \cos(x))(x^2 + 1) - (2x^2 + 1) \sin(x)}{(x^2 + 1)^2}$$

$$= \frac{[4x \sin(x) + (2x^2 + 1) \cos(x)](x^2 + 1) - (4x^3 + 2x) \sin(x)}{(x^2 + 1)^2}$$

$$c) f(x) = (x^2 - 1) \sec x \tan x \Rightarrow f'(x) = (x^2 - 1)' \sec x \tan x + (x^2 - 1) \sec x \tan^2 x + (x^2 - 1) \sec^3 x$$

$$f'(x) = 2x \sec x \tan x + x^2 - 1 \sec x \tan^2 x + (x^2 - 1) \sec^3 x$$

$$\text{si } (g \circ f)(x) = g(f(x))$$

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$$

Regla de la cadena $\rightarrow$

$$\text{Ej: } f(x) = \sin(2x^2 + 1) \Rightarrow f'(x) = \cos(2x^2 + 1) \cdot 4x = 4x \cos(2x^2 + 1)$$

NP $\geq 4,95$ para eximirse de la evaluación "examen"
Solamente reingresar en caso de faltar a una prueba [de carácter global]
Echó el examen $< 75\%$ asistencia

Unidad 0: Repaso de derivadas.

Derivada $\rightarrow$ la pendiente de la recta tangente a una curva en un punto dado.

Si $h \rightarrow 0$, se obtiene la pendiente de la recta tangente en el punto X.
$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x) = \frac{dy}{dx} = \frac{df(x)}{dx}$$

3B
GRAPHITE
ARTEL

Ejemplo. $f(x) = \sqrt{x}$
 $f(x+h) = \sqrt{x+h}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{(x+h - x)}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}$$

$$f(x) = x^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2} \cdot x^{\frac{1}{2} - 1}$$

$$= \frac{1}{2} \cdot x^{-\frac{1}{2}}$$

$$= \frac{1}{2} \cdot \frac{1}{x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}$$

Table de derivées trigonométriques.

$$f(x) = \sec(x) \Rightarrow f'(x) = \tan(x)$$

$$f(x) = \cos(x) \Rightarrow f'(x) = -\sin(x)$$

$$f(x) = \tan(x) \Rightarrow f'(x) = \sec^2(x)$$

$$f(x) = \sec(x) \Rightarrow f'(x) = \sec(x)\tan(x)$$

$$f(x) = \csc(x) \Rightarrow f'(x) = -\csc(x)\cot(x)$$

$$f(x) = \cot(x) \Rightarrow f'(x) = -\operatorname{cosec}^2(x)$$

$$f(x) = \arcsin \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \arccos \Rightarrow f'(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \arctan(x) \Rightarrow f'(x) = \frac{1}{1+x^2}$$